

DEVOPS - SECURITY - CLOUD COMPUTING STARTING

FEB 07 2026

Important reminders

1. Classes starts with prayers
2. Classes are recorded
3. Videos and lectures are posted on telegram

Basic Linux commands

FILE AND NAVIGATION COMMANDS

Commands are space and case sensitive

wsl—install # wsl - - install

prof prince

#princeprince

- 1.Touch command [touch] - the touch command creates a file on the system

A # touchprince

B #touch prince

Exmples

touch demo

#touch finance

touch .demo

#touch .finance

2. Confirm our files with list command [ls]

LS , or ls or Ls

3. Listing options

ls -a [list files plus hidden files and directories]

ls -l

ls -lt

ls -ltr

4. Clear [clear] - clear your screen

5. History [list all executed commands]

DIRECTORIES COMMANDS

Directories are also called folders

Command	Description	Examples
mkdir	to create folders / directories	mkdir departments mkdir countries mkdir vehicles
cd	change directory	cd finance cd students
pwd	print working directory present working directory	pwd
cd ~	go back to the home directory	cd ~
mv	rename directory	mv students liontech-students
cp	copy files and directories into other directory	cp demo.txt students

rmmdir	delete a directory	rmmdir canada
FILE EDITING IN LINUX USING COMMAND EDITORS		
Vi editor		
COMMAND	EXPLANATION	EXAMPLES
vi	vi is a command line editor create a file called fruits and then access the and type in ur favorite fruits	vi fruits touch fruits vi fruits
Editor modes 1. command mode [readly only access and cat write] 2. insert mode [we can modify the file] save and quit 1. :wq! quit without saving 2. :q!		

Sunday, February 8, 2026

User management in linux or other os

- Access control
- Authorization
- Authentication

John give his car to James to go get his brief case from the car

- Access control

Car key - offers authentication

Authorization: bring suit case .

Cloud: IAM

IAM

I = IDENTIFY [who , what departments , manager , job title]

A = ACCESS [all files]

M = MANAGEMENT

CIA TRIAD [FILE PERMISSIONS]

C= CONFIDENTIALITY

I = INTEGRITY

A = AVAILABILITY

ELVIS - let in John = access

- Authentication
- John is able to cook /user the kitchen = authorization

Types of users in linux

1. Root user is also called - super user - they have the overriding previllage on the system.
2. Normal user
3. System

User and group management commands

Command	Description	Examples
whoami	tells who u are and displays ur username	whoami
id	displays user characties uid [userid] gid [group id]	id
who	list the user and processes linked with the user	
useradd adduser	these commands are used to add users to the system useradd adduser sudo to assume admin previllage sudo = super doer if u wantv to become root user sudo su -	adduser prof sudo adduser nico sudo adduser charma sudo adduser john sudo adduser mary
su	switch user navigating between accounts	su - john su - mary

10 MINUTES BREAK

Commands	Explanations	Examples
passwd	changing password # only admins can change password of other users	Passwd mike
	User management files /etc/shadow - Contains all details of ur users /etc/password	
Group management in linux		
commands	description	examples
groupadd	add a new group	sudo groupadd finance sudo groupadd students
usermod	add users to group	sudo usermod -aG finance mike sudo usermod -ag students uche
deluser	remove users from specific group	sudo deluser mike hr
userdel	remove/delete a user account	sudo userdel demo-user

FILE PERMISSIONS IN LINUX

TYPES OF USER	PERMISSIONS	IONOTES
Owner R = read W = write X = execute Group R = read W = write X = execute Others Read , no write permission , they execute R = - = no permission X = EXECUTE	R = READ W X - D rwx rwx r-x	DEFAULT IONOTE NUMBERS R = 4 W = 2 X = 1 Owner = wrx = full permission If the owner hs just x = 1
deault permissions with files owner: rw- = read, write , no executable permissions group: rw- other : r - - - = no permission 0 permission		

Exercise

Create a file called cars.txt and perform the following

1. Grant the following permission to the file
 - a. Owner has no permission
 - b. Group has no permission
 - c. Others have no permission

USERS	R	W	X	TOTAL PERMISSION
OWNER	0	0	0	0+0+0 = 0
GROUP	0	0	0	
OTHERS	0	0	0	

To change permission we use the chmod command

```
# chmod 000 cars.txt
```

b. now lets implement the following security protocol

1. The owner should be able to read

```
OTHER    R = 0 W = 0  = X 0
GROUP    R = 0 , W = 0  X = 0
OWNER    R=4 , W = 0 X =0
```

```
Chmod 400 cars.txt
```